INCH-POUND
MIL-DTL-16377/16B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/16A(SH)
18 March 1982

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR DESKS, 8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 141.2, 150.2, 141.2SSL, AND 150.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 7 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

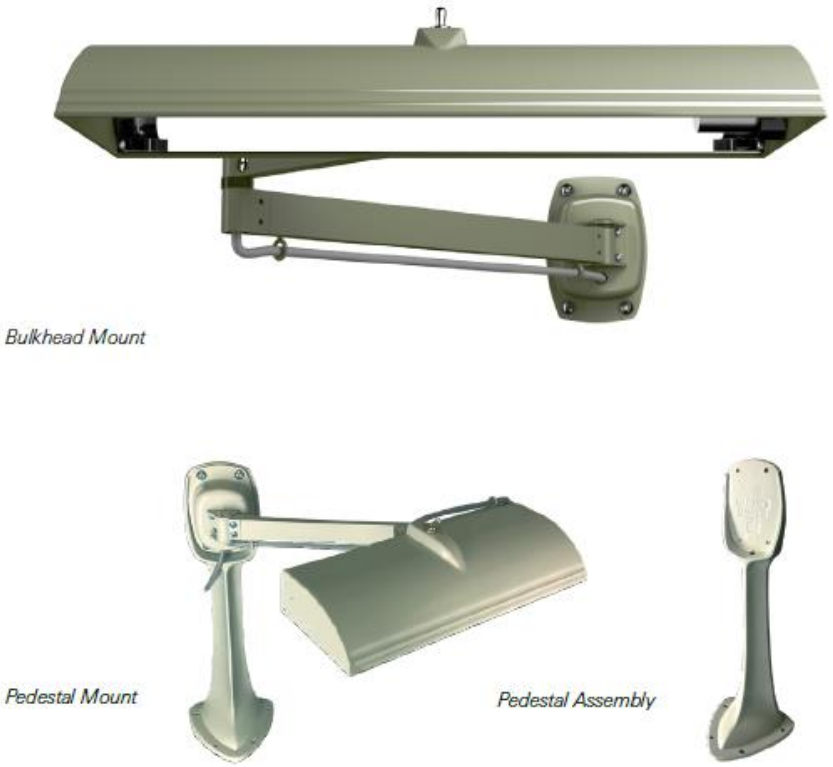


FIGURE 1. Typical view of symbols 141.2 and 150.2.

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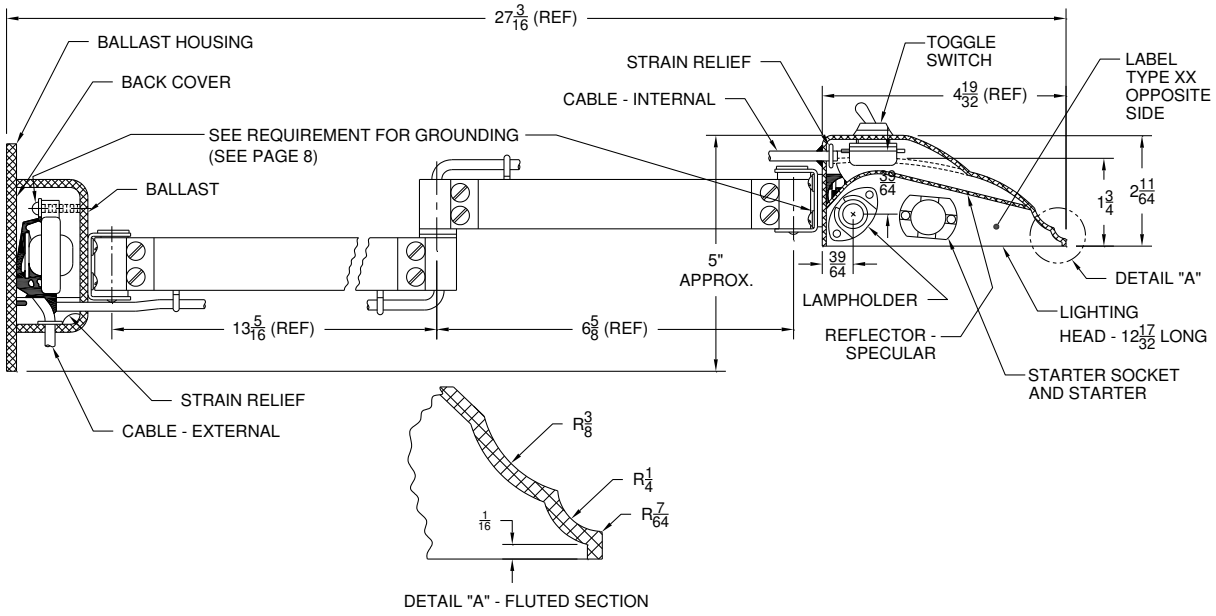


FIGURE 2. Typical section for symbols 141.2 and 150.2.

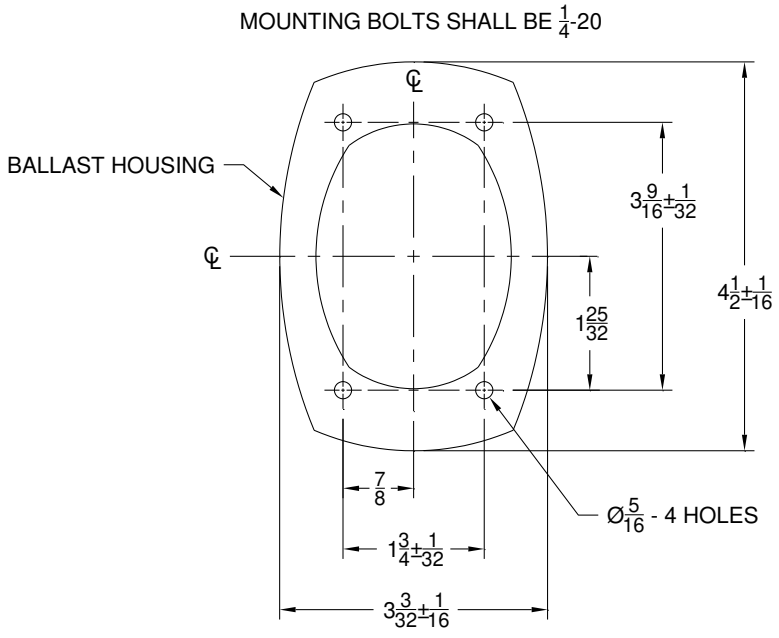


FIGURE 3. Mounting dimensions for symbol 141.2.

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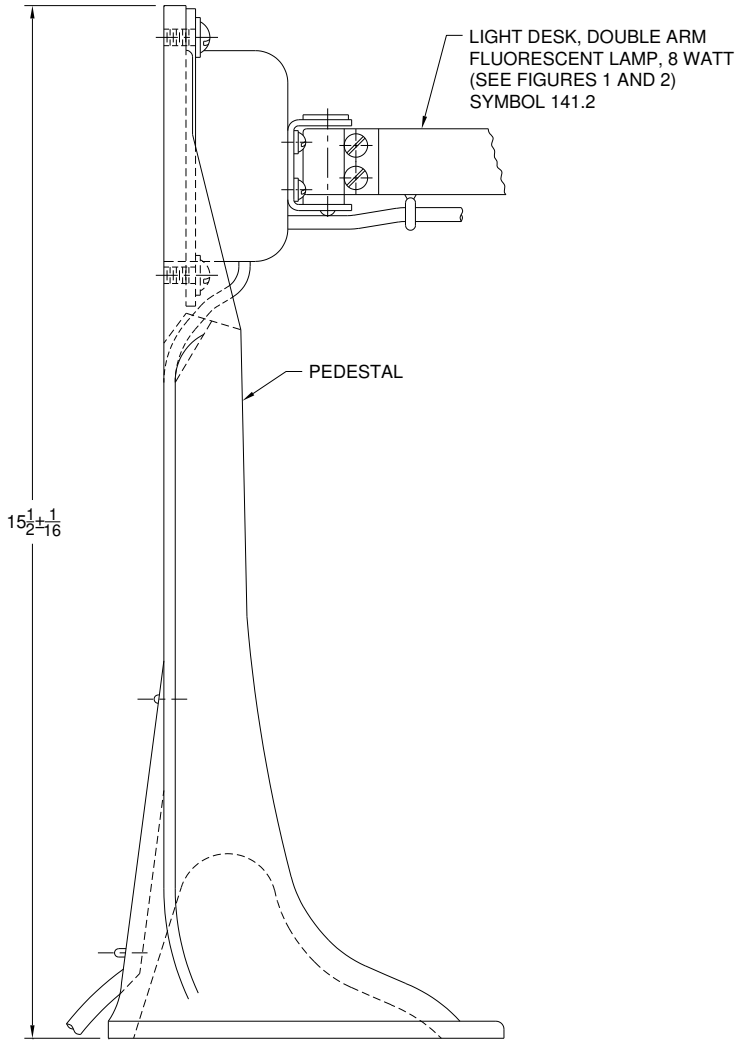


FIGURE 4. Pedestal for symbol 150.2.

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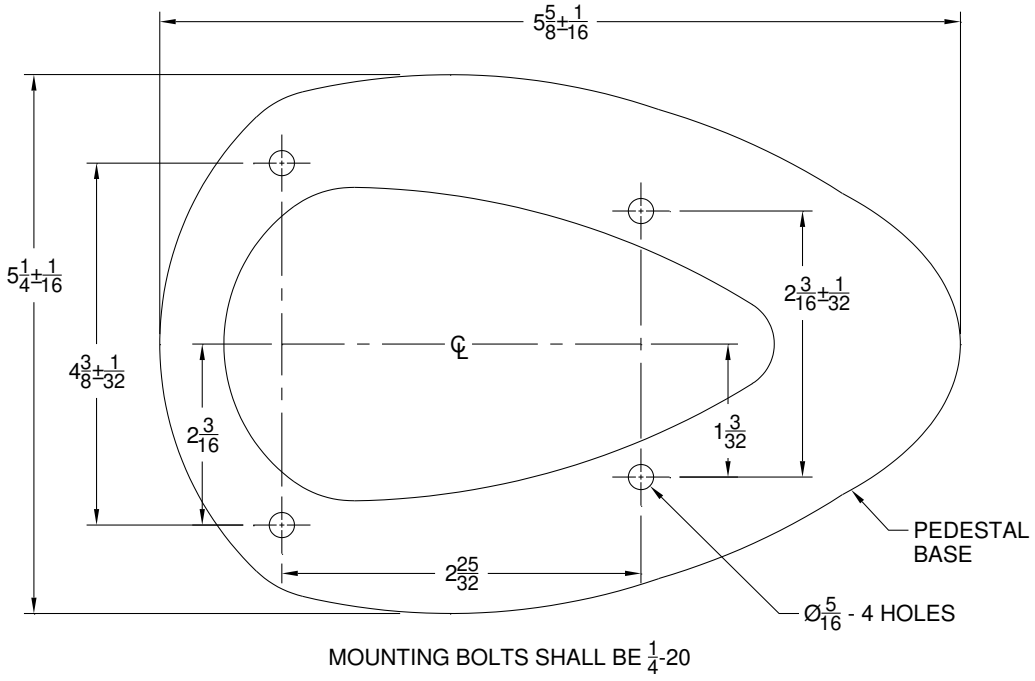


FIGURE 5. Mounting dimensions for symbol 150.2.

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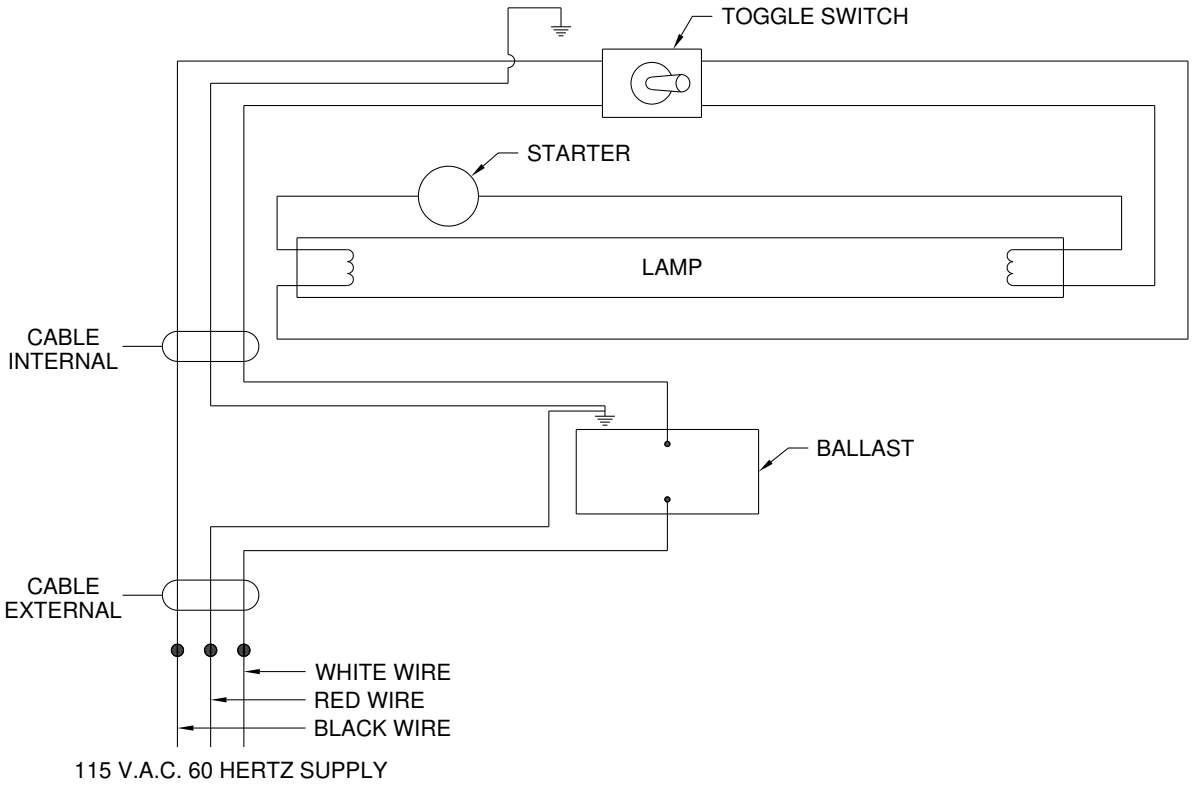
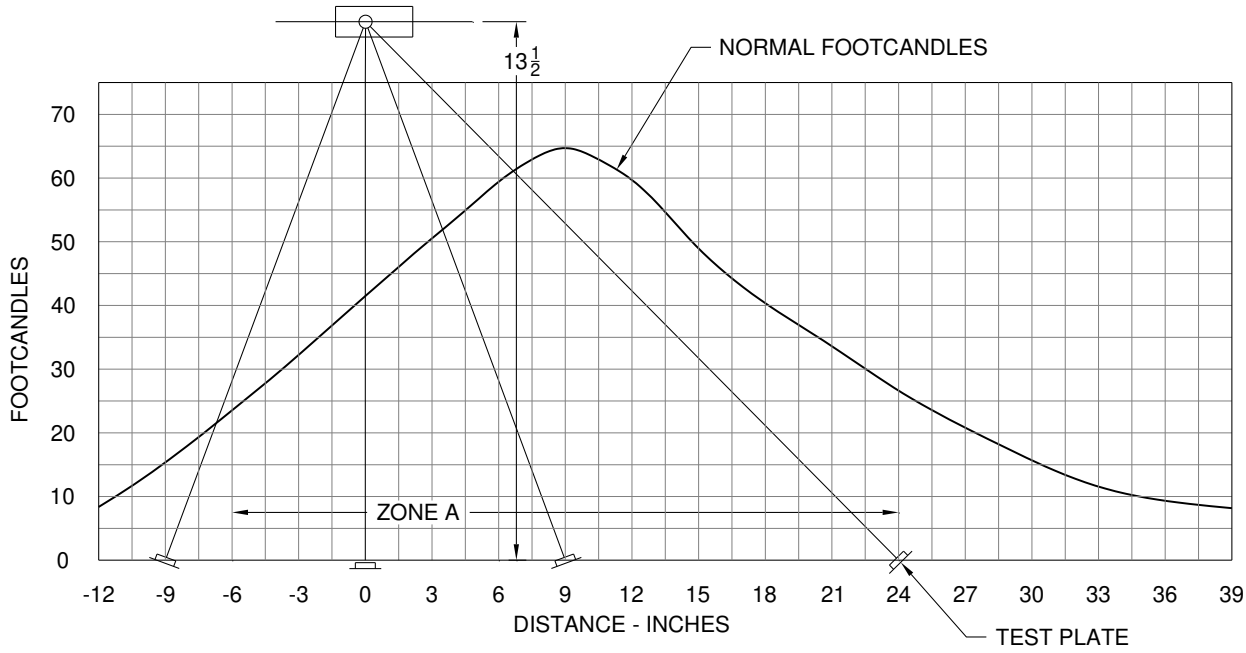


FIGURE 6. Wiring diagram for symbols 141.2 and 150.2.

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Photometric test data:

Distance: 13½ inches

Voltage: 115 volts AC, 1 phase

Ambient temperature: 70 °F

Lamp and ballast furnished with fixture shall be used when testing.

FIGURE 7. Footcandle distribution curves for symbols 141.2 and 150.2.

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FLUORESCENT REQUIREMENTS

CLASSIFICATION

Type: I
Class: 1

SYMBOL NUMBERS

141.2: Light, desk double arm, bulkhead mounted with one 8-watt fluorescent cool white lamp.
150.2: Light, desk, double arm with pedestal and one 8-watt fluorescent cool white lamp.

DESIGN

Electrical rating:	8 watts, 120 volts, 60 hertz.
Dimensions and configuration:	
Symbol 141.2:	See figures 1, 2, and 3.
Symbol 150.2:	See figures 1, 2, 4, and 5.
Distribution curves:	Footcandle (see figure 7).
Typical section:	See figure 2.
Cable:	Internal and external cable shall be type TX3 in accordance with MIL-DTL-24640/2 . For internal and external cable, the ground leads inside ballast housing shall be terminated with crimp-type terminals and attached under the screw and washer. For internal cable, the ground leads inside the lighting head assembly shall be terminated with crimp-type terminals and connected under the round head machine crew and lock-washer. 31 inches of internal and 10 feet of external cable shall be provided.
Lighting head and supporting arms:	The lighting head shall be designed to house the necessary parts and shall be of a shape that will accommodate the filter assembly shown on MIL-F-16377/48 symbols 121.1 and 121.2. The lighting head shall be made of cast aluminum; minimum thickness shall be 3/32 inch. The supporting arms shall be made of extruded aluminum; minimum thickness shall be 0.024 inch. Swivel pins shall be extended nylon and hollow corrosion resistant steel tubing. Swivel brackets and wire loops shall be aluminum. Swivel joints shall hold the arm firm at all adjustments. Swivel action shall be smooth for all adjustments in the horizontal plane.
Pedestal (symbol 150.2 only):	See figure 4. Material shall be cast aluminum.
Wiring diagram:	See figure 6.
Reflector:	Specular, prefinished lighting sheet.
Weight:	
Symbol 141.2:	3 pounds, 9 ounces maximum.
Symbol 150.2:	6 pounds, 0 ounces maximum.
Ballast housing:	Aluminum, 1/8 inch minimum thickness.
Back cover:	Laminated phenolic material.
Strain relief:	Strain relief for cables shall be provided as shown on figure 2.
Designation and markings:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1 and 3: (a) One type I. (b) One type IV. The information to be shown on the type IV label shall be "USE LAMP NSN 6240-00-299-5546 ONLY."
Standard parts:	The standard parts shown in table I shall be used in the construction of the fixtures specified herein with no deviation.

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SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBERS

141.2SSL: Light, desk double arm, bulkhead mounted, SSL, white
150.2SSL: Light, desk double arm with pedestal, SSL, white

DESIGN

Electrical rating:	8 watts, 120 volts, 60 hertz.
Maximum dimensions:	See figures 2 and 4.
Wiring:	Shall maintain the same control scheme for light output as legacy fluorescent fixture.
Cable:	10 feet, TX in accordance with MIL-DTL-24640/2.
Photometric data:	See figure 7 of this sheet and 3.7.1 of MIL-DTL-16377. The minimum luminescence shall be not less than that shown on figure 7.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Weight:	
Symbol 141.2SSL:	3 pounds, 9 ounces maximum.
Symbol 150.2SSL:	6 pounds, 0 ounces maximum.
Designation and markings:	One type I label shall be furnished in accordance with MIL-DTL-16377.
Material:	Corrosion-resistant in accordance with MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examination and tests shall be as specified on MIL-DTL-16377 and as follows:

First article inspection:	See table II or III.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information for fluorescent fixtures are shown on table IV. These lights are designed to receive a red and white or blue and white filter, as shown on MIL-F-16377/48, symbols 121.1 and 121.2. The filters are not furnished with the light fixtures. These lights were formerly covered by NAVSHIPS drawing 9000-S6401-73811 and MS16719.

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TABLE I. Standard parts for type I fixtures.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry Number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	---
Lampholder	UL 542	6250-00-377-8590	Supplement SA
Startersocket	UL 542	6250-00-295-2738	Supplement SA
Starter	ANSI C78.180	6250-00-299-5962	Commercial designation: FS 5
Ballast	M16377/45-001	6250-00-295-2735	---
Toggle switch DPST	MS16377/45-1	5930-00-719-9659	---
Red and white filter	M16377/48-121.1	6230-00-548-2040	Not furnished
Blue and white filter	M16377/48-121.2	---	Not furnished

TABLE II. First article and comparison inspection for type I fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Shock	3.5.12	4.8.7	1/
Vibration	3.5.15	4.8.8	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	---
Grounding circuit	3.6.13.3	4.8.1	---
Load test	2/	2/	2/
Light output	3.7.2	4.8.8.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	---

NOTES:

1/ Lamp breakage or lamp fallout during this test does not constitute failure.

2/ With the desk light installed in the regular mounting position, a vertical load of 15 pounds applied to the swivel joints adjacent to the head assembly shall deflect this joint not greater than ¾ inch.

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TABLE III. First article and comparison inspection for complete type III fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Power interface	3.6.18	4.8.21	---
Shock	3.5.12	4.8.7	Grade B 1/
Vibration	3.5.13	4.8.8	1/
Brightness	3.7.5	4.8.6.4	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Photometrics	3.7.1	4.8.6	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Failsafe circuit design	3.6.17	4.8.22	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

TABLE IV. Part number and general information.

Military Part Number M16377/16-	Item Name	National Stock Number (NSN)
141.2	Complete light, symbol 141.2	6230-00-067-4882
150.2	Complete light, symbol 150.2: Symbol 141.2 and pedestal	6230-00-067-4882 6230-00-336-5876
141.2SSL	Complete light fixture, symbol 141.2SSL	---
150.2SSL	Complete light, symbol 150.2SSL	---
001	Pedestal	6230-00-336-5876

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6230-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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